Software Requirements Specification (SRS)

Change Management Software

Project	Change Management Software (CMS)
Version	2.0 (as built)
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Date	7 November 2026
Status	Final: describes the release built in Sprints 1 and 2 (feature freeze 7 Nov 2026)

Revision History

Version	Date	Author	Change summary	Approval
1.0	28 October 2026	Team 4	Initial project-specific draft	Team
2.0	7 November 2026	Team 4	Updated to the implemented system: as-built technology, design and test results; implementation status and deviations recorded	Team

Approvals

Role	Name	Signature / Email	Date
Team lead	Vivaan Atharva		
Course instructor / TA	To be filled by instructor	Pending	

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1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for the Change Management Software (CMS), an issue management application in which users raise change requests, reviewers approve and assign them to engineers, engineers resolve them, and every request is tracked through to closure. It is the reference for implementation, testing, and project evaluation. Version 2.0 keeps every requirement as originally specified and adds what the delivered release (end of Sprint 2) implements: section 9 traces each requirement to its Jira story and automated tests, and section 10 lists what was not built.

1.2 Scope

CMS supports creating change requests, reviewing and approving them, assigning them to engineers, recording progress, comments and resolutions, verifying and closing completed work, reopening work that was not resolved correctly, and reporting on open, overdue and closed requests. The initial project is a software prototype for a single organisation or team. It does not include source-code repository integration, CI/CD deployment automation, email/chat integrations, or time and billing tracking unless added and approved later.

1.3 Audience

Student developers, testers, users acting as requesters, reviewers and engineers, project evaluators, and the teaching assistant.

1.4 Definitions

Term	Definition
CMS	Change Management Software
Change request (CR)	A recorded request for a change, fix or enhancement, with its own ID, details, status, priority and history
Workflow	The defined set of CR statuses and the allowed transitions between them
Assignee	The engineer currently responsible for working on a CR
Resolution	The engineer's summary of what was done to address a CR
SLA target	The target number of days to resolve a CR, configured per priority level
Overdue	A CR that is not Resolved or Closed after its due date has passed
FR / NFR	Functional Requirement / Non-functional Requirement
RBAC	Role-Based Access Control

2. Overall Description

2.1 Product Perspective

CMS is a web application with a user interface, an application/service layer, and a persistent database. Users enter change request details; the application validates and stores them, enforces the status workflow, records history, flags overdue work, and presents dashboards and reports.

2.2 Major Product Functions

- Authenticate users and apply role-based permissions.
- Create, view, edit, search and filter change requests.
- Review change requests and approve or reject them with a recorded reason.
- Assign and reassign approved change requests to engineers.
- Record work progress, comments and a resolution summary.
- Verify resolved work and close it, or reopen it with a reason.
- Maintain a complete status and assignment history for every change request.
- Flag change requests that are overdue against their priority's SLA target.
- Show in-app notifications for assignments, status changes and comments.
- Generate status, workload, ageing and resolution-time reports.

2.3 User Roles and Characteristics

Role	Needs / permissions
Administrator	Manage users and roles; configure components, priorities and SLA targets; view all change requests, reports and audit history.
Change manager / reviewer	Review new change requests; approve or reject them; assign and reassign engineers; verify and close resolved requests; view reports.
Engineer	View change requests assigned to them; update progress; add work notes and comments; mark requests as resolved.
Requester	Create change requests; view and comment on their own requests; confirm a resolution or ask for the request to be reopened.
Viewer / evaluator	Read dashboards, change requests and reports without changing data.

2.4 Operating Environment

As built, CMS is a Python 3.12 web application using Flask 3 with server-rendered Jinja templates and Bootstrap 5.3, Flask-SQLAlchemy over a SQLite database file, and Flask-Login for sessions. It runs in any modern desktop browser. The repository README gives setup steps for macOS and Windows; every pull request is tested by GitHub Actions on Ubuntu, Windows and macOS.

2.5 Constraints and Assumptions

- Users enter change request details accurately; the prototype does not verify that the described change was actually made in any external system.
- Each change request has exactly one current status and at most one current assignee.
- Only active users with the Engineer role can be assigned a change request.
- SLA targets are configurable or use documented defaults per priority.
- The system serves one organisation in the initial release unless multi-team support is explicitly implemented.
- All dates and due-date calculations use the organisation's configured local time zone.
- The implementation and selected technologies must match the actual repository; record deviations in the final demo/documentation.

3. External Interface Requirements

3.1 User Interface

- Dashboard showing open change requests by status, requests assigned to the signed-in user, and overdue requests.
- Change request detail page showing fields, current status, assignee, comments and full history timeline.
- Forms with required-field validation and clear error messages.
- Change request list with search, sort, and filters for status, priority, assignee, requester and component.
- Status and priority indicators that do not rely on colour alone.

3.2 Software Interfaces

- Application services expose operations for change request, workflow, assignment, comment, notification and report workflows.
- The database stores users, roles, change requests, status history, comments, notifications, settings and audit events.
- Optional email, chat or source-control integrations are outside the initial scope unless implemented and documented.

3.3 Communications

For local development, browser-to-server communication may use localhost. Any deployed environment should use HTTPS, protect credentials and session tokens, and avoid exposing the database directly to end users.

4. System Features / Functional Requirements

Req ID	Requirement (shall)	Priority	Acceptance criteria
CMS-F-001	The system shall allow an authorized user to sign in and sign out.	High	Valid credentials create a session; invalid credentials are rejected.
CMS-F-002	The system shall enforce role-based permissions for administrative, review, assignment and resolution actions.	High	A user without permission cannot access or execute a restricted action.
CMS-F-003	The system shall allow an authenticated user to create a change request with title, description, type, priority and affected component.	High	A valid CR is saved with a unique ID, status Submitted, and the creator recorded as requester.
CMS-F-004	The system shall validate that required CR fields are present and that type, priority and component are allowed values.	High	Missing or invalid values are rejected with a clear message and nothing is saved.
CMS-F-005	The system shall allow authorized users to view, search, sort and filter change requests by status, priority, assignee, requester, component and date.	High	Search/filter results match the selected criteria.
CMS-F-006	The system shall allow the requester or a reviewer to edit CR details while the CR is not Closed.	Medium	Edits persist and appear in the CR history; edits to a Closed CR are rejected.
CMS-F-007	The system shall enforce the defined workflow: Submitted, Under Review, Approved or Rejected, Assigned, In Progress, Resolved, Closed, and Reopened.	High	Only allowed transitions succeed; e.g. Submitted to Closed is rejected.
CMS-F-008	The system shall allow a	High	Approval moves the CR to

	reviewer to approve or reject a CR, and shall require a reason when rejecting.		Approved; rejection without a reason is refused.
CMS-F-009	The system shall allow a reviewer to assign or reassign an approved CR to an active engineer.	High	Assignee is updated, CR moves to Assigned, and the assignment is recorded in history.
CMS-F-010	The system shall allow the assigned engineer to start work and add work notes; users with access to a CR shall be able to add comments.	High	Status changes to In Progress; notes and comments are saved with author and timestamp.
CMS-F-011	The system shall allow the assigned engineer to mark a CR as Resolved and shall require a resolution summary.	High	CR moves to Resolved only when a resolution summary is provided.
CMS-F-012	The system shall allow the requester or a reviewer to close a Resolved CR, or reopen it with a reason.	High	Closing sets status Closed with closure date; reopening returns the CR to the assignee with the reason recorded.
CMS-F-013	The system shall record every status change and assignment change with previous value, new value, acting user and timestamp.	High	The CR history timeline lists every transition in order.
CMS-F-014	The system shall calculate a due date for each CR from its priority's SLA target and flag CRs that are past due and not Resolved or Closed.	Medium	A CR past its due date appears in the overdue view; a resolved CR does not.
CMS-F-015	The system shall create in-app notifications for the relevant users when a CR is assigned, changes status, or receives a comment.	Medium	The affected user sees an unread notification linked to the CR.
CMS-F-016	The system shall show a dashboard with CR counts by status and priority, the signed-in user's assigned CRs, and overdue CRs.	Medium	Dashboard counts match the stored CR data.
CMS-F-017	The system shall generate reports of open CRs by status, priority, assignee and component for a current snapshot.	Medium	Report totals reconcile with the CR data for the selected scope.
CMS-F-018	The system shall generate ageing and resolution-time reports for a selected date range.	Medium	The report shows each CR's age or time-to-close and the average for the range.
CMS-F-019	The system shall record an audit event for important changes to CRs, users, roles and settings.	High	Audit entries identify action, timestamp, and acting user.
CMS-F-020	The system shall allow an administrator to configure components, priority levels and SLA targets.	Medium	New settings apply to CRs created or re-prioritised afterwards.
CMS-F-021	The system shall display validation and operation errors without exposing stack traces or sensitive data.	High	User sees a useful message while technical details remain in protected logs.

5. Non-functional Requirements

Req ID	Requirement	Category	Acceptance / measurement

CMS-NF-001	For a dataset of up to 10,000 change requests on the project test machine, common CR searches, list views and dashboard loads should complete within 2 seconds for at least 90% of measured requests.	Performance	Record timings across at least 20 repeated requests; 90% meet target.
CMS-NF-002	The application shall preserve committed change requests, history, comments and settings across normal application restarts.	Reliability	Restart application and verify previously committed records remain intact.
CMS-NF-003	Passwords shall not be stored in plaintext; access to protected actions shall require authentication and authorization.	Security	Inspect stored user data and test unauthorized requests.
CMS-NF-004	The user interface shall provide consistent navigation, readable labels, validation feedback, and keyboard-operable core forms where supported by the framework.	Usability / Accessibility	Complete defined usability checklist for create, assign, resolve and close workflows.
CMS-NF-005	The codebase shall separate UI/routes, business logic/services (including the workflow rules), and data persistence sufficiently to support unit testing and maintenance.	Maintainability	Workflow and assignment rules can be unit-tested without launching the full UI.
CMS-NF-006	The application shall provide a documented setup procedure and run on the team's supported development environment.	Portability	A team member follows README setup steps on a clean environment and launches the application.
CMS-NF-007	When two users update the same CR at the same time, the system shall not silently overwrite either change.	Integrity	The second conflicting update is rejected with a message asking the user to reload.

6. Security Objectives and Requirements

6.1 Security Objectives

- Protect change request and user-account data from unauthorized viewing or modification.
- Preserve the integrity and traceability of status changes, assignments and resolutions so that every decision can be investigated.

6.2 Security Requirements

Req ID	Requirement	Priority	Acceptance / test
CMS-SR-001	The system shall require authentication before allowing any CR to be created or changed.	High	Unauthenticated requests to protected operations are rejected.
CMS-SR-002	The system shall enforce role-based authorization on the server/service layer, not only by hiding UI controls.	High	Direct requests by a role without permission are denied.
CMS-SR-003	The system shall store passwords using a suitable salted password-hashing mechanism and shall never log passwords.	High	Storage and log inspection show no plaintext passwords.
CMS-SR-004	The system shall validate and	High	Malformed or script-containing

	normalize user-supplied values on the server side and escape user-entered text when displaying it.		input is rejected or rendered harmlessly.
CMS-SR-005	The system shall enforce record-level access: engineers may update only CRs assigned to them and requesters may edit only their own CRs.	High	Updating another user's CR by direct request is denied.
CMS-SR-006	The system shall record audit events for workflow transitions, assignments and privileged changes without including passwords or other secrets.	High	Audit history records action and actor while excluding secret values.

7. Quality Attributes and Acceptance Tests

Acceptance exit criteria: all high-priority functional requirements have been implemented and tested; no open critical defect remains; every CR's history is consistent with its current status; and each implemented requirement is linked to at least one test case in the traceability matrix.

Quality attribute	Acceptance test
Correctness	Walk CRs through every allowed transition and confirm invalid transitions are rejected and history matches current status.
Security	Test sign-in, role restrictions, record-level access, input validation, and audit-log content.
Performance	Measure dashboard and CR search against CMS-NF-001.
Usability	Ask a tester to create a CR, approve and assign it, resolve it and close it using the UI alone.
Maintainability	Run unit tests for workflow, assignment and SLA services and verify results are repeatable.
Portability	Follow documented setup steps in a clean virtual environment.

8. UML Use-Case Models

Use-case diagram 1 — Change request lifecycle. Actors: Requester, Change Manager/Reviewer and Engineer. Use cases: Sign in; Create change request; Search/view change requests; Comment on change request; Review and approve/reject; Assign/reassign engineer; Start work and add work notes; Resolve change request; Verify and close; Reopen change request. Requesters create, comment on and verify their own requests; reviewers approve, assign and close; engineers work on and resolve requests assigned to them.

Use-case diagram 2 — Administration and reporting. Actors: Administrator and Change Manager/Reviewer. Use cases: Manage users/roles; Configure components, priorities and SLA targets; View dashboard; View open-CR report; View ageing and resolution-time report; View overdue CRs; Review audit history. Engineers and requesters may view the dashboard and notifications but cannot manage users or change system-wide settings.

9. Requirements Traceability Matrix (RTM)

Status at the feature freeze (7 Nov 2026). "Pass" means the requirement is implemented and its automated tests pass in GitHub Actions on Ubuntu, Windows and macOS (337 tests, 0 failures). The full matrix, with the name of every automated test, is kept in the repository at docs/RTM.md.

Req ID	Short requirement	Jira story	Test case(s)	Status
CMS-F-001	Sign in/out	CMS-2	TC-AUTH-01	Pass
CMS-F-002	Role-based permissions	CMS-2, 5, 7	TC-AUTH-02	Pass
CMS-F-003	Create change request	CMS-3	TC-CR-01, TC-CR-02	Pass
CMS-F-004	Validate CR fields	CMS-3	TC-VAL-01	Pass
CMS-F-005	View, search, sort, filter	CMS-4	TC-CR-01	Pass (no date-range filter)

CMS-F-006	Edit CR until closed	CMS-11	TC-CLOSE-01	Pass
CMS-F-007	Enforce workflow	CMS-5	TC-WF-01, TC-WF-02	Pass
CMS-F-008	Approve / reject with reason	CMS-6	TC-WF-03	Pass
CMS-F-009	Assign / reassign engineer	CMS-7	TC-ASSIGN-01, TC-ASSIGN-02	Pass
CMS-F-010	Start work; notes and comments	CMS-8, CMS-10	-	Partial: start work only
CMS-F-011	Resolve with summary	CMS-8	TC-RES-01	Pass
CMS-F-012	Close / reopen with reason	CMS-9	TC-CLOSE-01, TC-REOPEN-01	Pass
CMS-F-013	Status and assignment history	CMS-5, 7, 11	TC-HIST-01	Pass
CMS-F-014	Due dates and overdue flag	CMS-13, CMS-12	TC-SLA-01, TC-SLA-02	Pass
CMS-F-015	In-app notifications	CMS-14	-	Not implemented
CMS-F-016	Dashboard	CMS-12	TC-REP-01 (dashboard part)	Pass
CMS-F-017	Open-CR reports	-	TC-REP-01	Not implemented
CMS-F-018	Ageing / resolution-time reports	-	TC-REP-02	Not implemented
CMS-F-019	Audit events	CMS-5, 7, 11	TC-AUDIT-01	Partial
CMS-F-020	Configure components, priorities, SLAs	CMS-13	-	Partial: SLA days by setting
CMS-F-021	Friendly errors	CMS-1, CMS-3	TC-CR-02	Pass
CMS-NF-001	Performance target	-	TC-PERF-01	Not measured
CMS-NF-002	Data survives restart	CMS-1	TC-RESTART-01	Pass
CMS-NF-003	Secure credentials	CMS-2	TC-SEC-01	Pass
CMS-NF-004	Consistent, accessible UI	All UI stories	Manual check	Pass
CMS-NF-005	Layered, testable code	CMS-1, CMS-5	Service-level tests	Pass
CMS-NF-006	Documented setup, runs on team OSes	CMS-1	CI on 3 OSes	Pass
CMS-NF-007	No silent overwrite on concurrent edits	-	TC-CONC-01	Not implemented
CMS-SR-001	Login required	CMS-2	TC-AUTH-02	Pass
CMS-SR-002	Server-side authorization	CMS-2, 5, 7	TC-AUTH-02	Pass
CMS-SR-003	Salted password hashing	CMS-2	TC-SEC-01	Pass
CMS-SR-004	Validation and output escaping	CMS-3	TC-VAL-01, TC-VAL-02	Pass
CMS-SR-005	Record-level access	CMS-4, 5, 11	TC-ASSIGN-03	Pass
CMS-SR-006	Audit without secrets	CMS-5, 7	TC-AUDIT-01	Partial

10. Implementation Status and Deviations

Summary: 25 of the 34 requirements are fully implemented and tested, 4 are partly implemented, and 5 were not built or not measured in this release. The full change-request lifecycle (create, review, approve or reject, assign, start, resolve, close or reopen) works end to end and is covered by automated integration tests.

Requirement	What was delivered	Reason / follow-up
CMS-F-010 (comments, work notes)	Engineers can start work; comments and work notes were not delivered (Jira CMS-10).	Not completed by the feature freeze. Future work: Comment model and timeline on the detail page.
CMS-F-015 (notifications)	Not delivered (Jira CMS-14).	Lower priority; dropped at the freeze. The dashboard work lists (Assigned to me, Waiting for review) partly cover the need.
CMS-F-017, CMS-F-018 (reports)	Not delivered. The dashboard shows counts by status and priority, work lists and overdue items.	Not planned in Sprint 1 or 2. Future work: report pages built on the dashboard queries.
CMS-F-019, CMS-SR-006 (audit)	Every status change, assignment and field edit is recorded with the acting user and time. Changes to users and settings are not audited.	There are no screens to change users or settings in this release.
CMS-F-020 (configuration)	SLA days per priority can be changed with the CMS_SLA_DAYS setting. Components and priorities	No admin settings screen in this release.

	are fixed in code.	
CMS-F-005 (date filter)	Search, sort (newest/oldest/priority) and filters for status, priority, assignee, requester, component and overdue.	No date-range filter.
CMS-NF-001 (performance)	Not measured.	Would need a 10,000-CR dataset and timing runs.
CMS-NF-007 (concurrent edits)	Not implemented; the last save wins.	Future work: version column and optimistic locking.
Viewer role (section 2.3)	Not implemented. The roles are Requester, Reviewer, Engineer and Admin.	Not needed for the demonstrated workflow.